



Stellenbosch

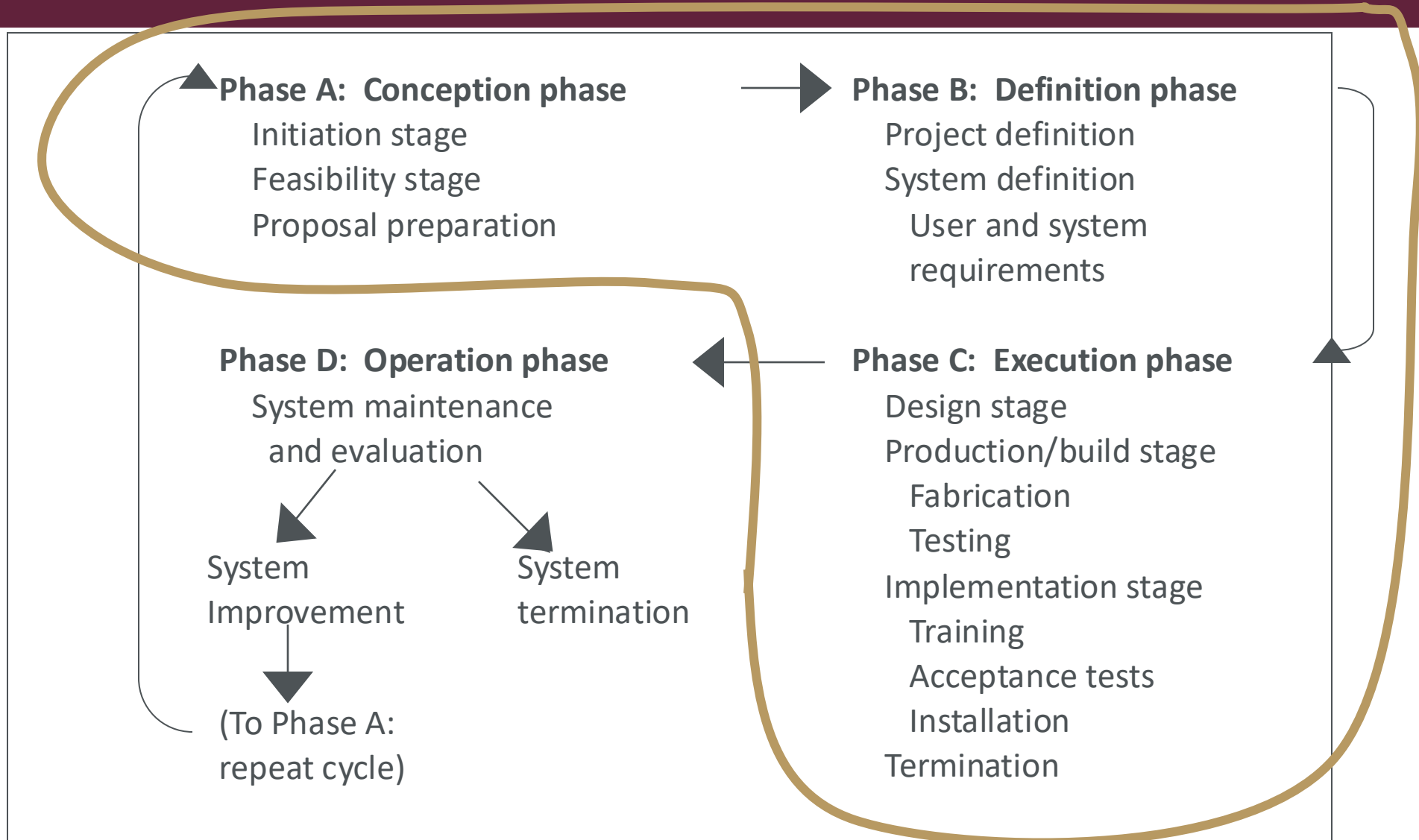
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CH03: PROJECT LIFE CYCLE AND PROJECT CONCEPTION

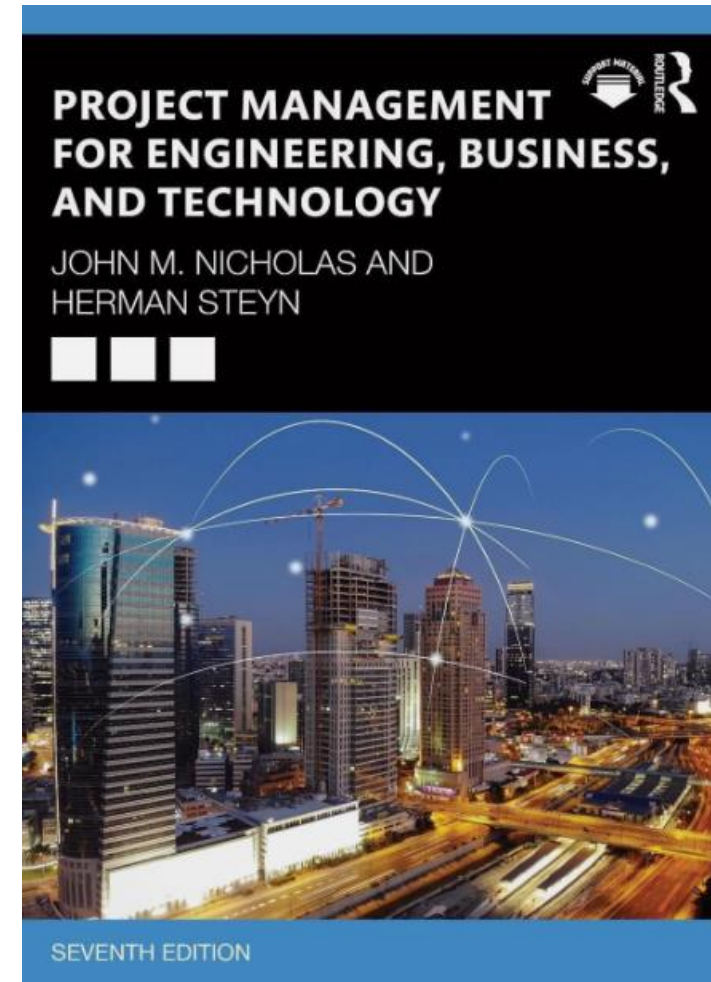
Prof Corné Schutte

Systems Development Cycle/Project Life Cycle



3.1 Project Life Cycle

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John M Nicholas
Herman Steyn



Life Cycle Stages: Natural Organisms

- All living organisms follow life-cycle stages

Conception

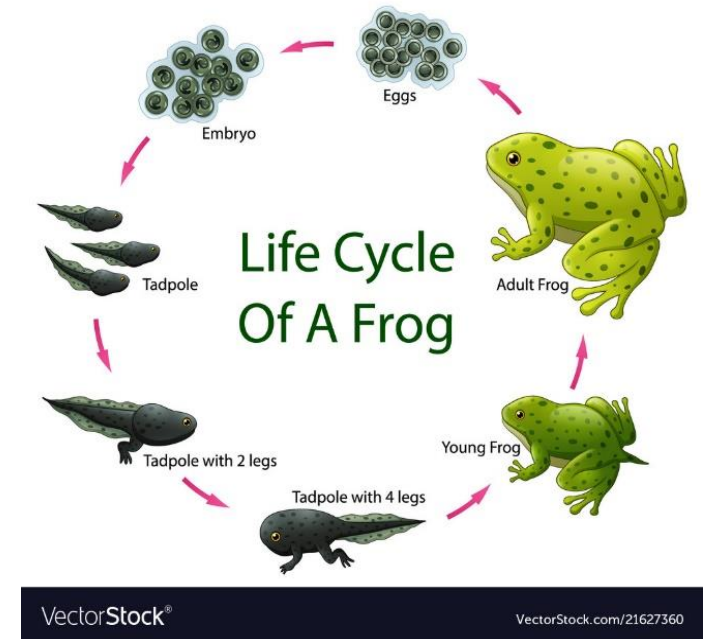
Birth

Growth

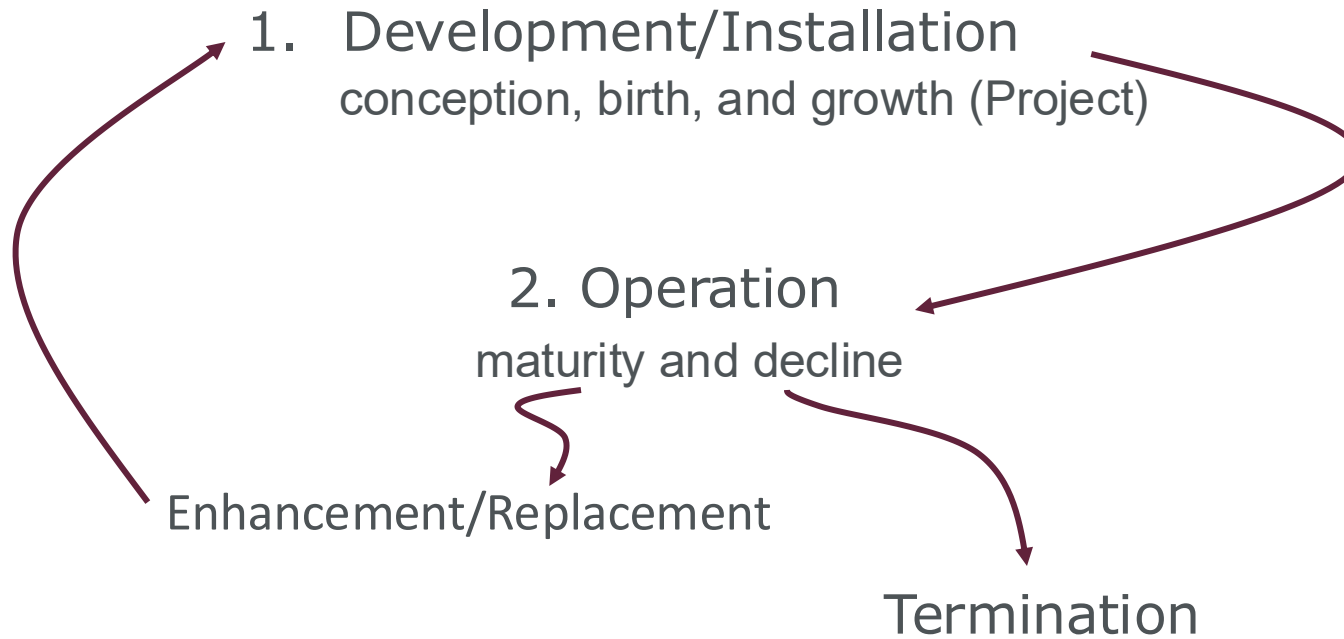
Maturity

Decline

Death



Life Cycle Stages: System Life Cycle



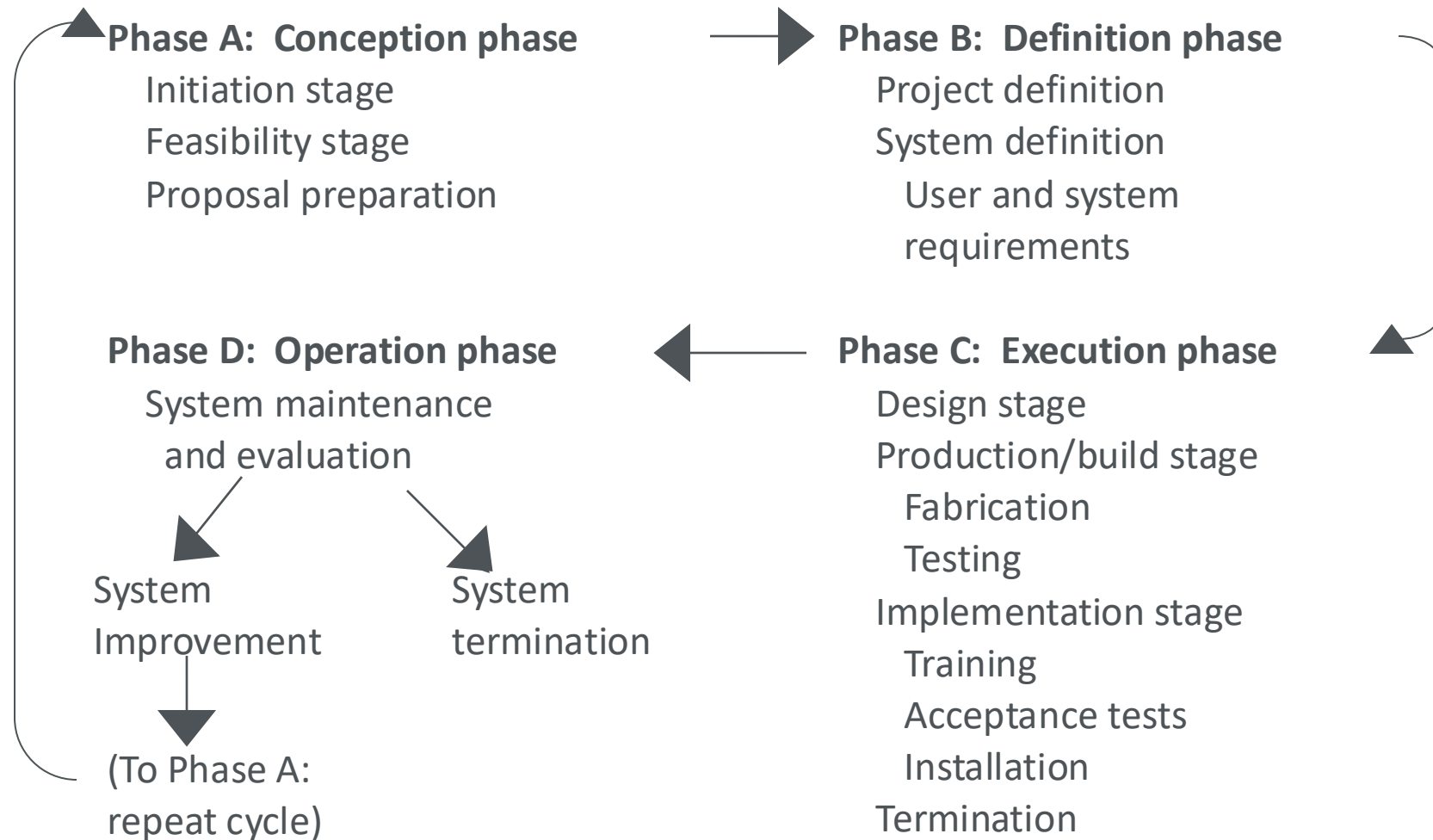
- For a human-made system **termination** is not inevitable.
- System is **kept “alive”** through enhancement/ replacement.
- Thus, *every* human-made system **begin** as a project and often **ends** with the start of a new project.

3.2 Systems Development Cycle

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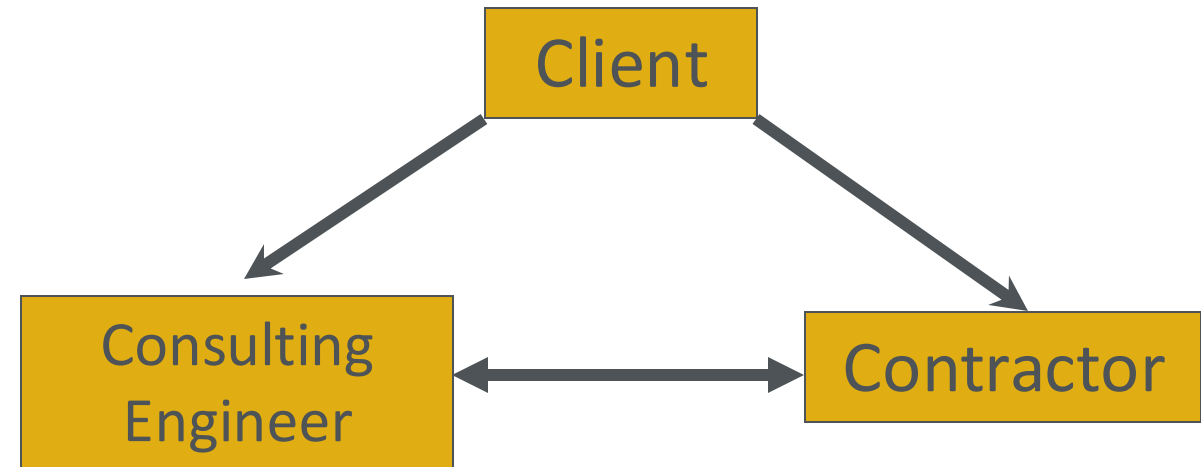


Systems Development Cycle or Project Life Cycle



Key Actors (Stakeholders) in SDC

- Client
 - Party **for whom** project is being done
 - **Pays** for project
 - Has needs and requirements **to be met**
- User/Customer
 - Party that operates or is **beneficiary** of project end-item
 - Might be same as customer, might not
 - If different, important to **differentiate** user from customer
- System Development Organization (SDO)
 - Party that **performs work** for customer
 - aka developer, contractor, consultant



Process: RFP, Proposal and Contract

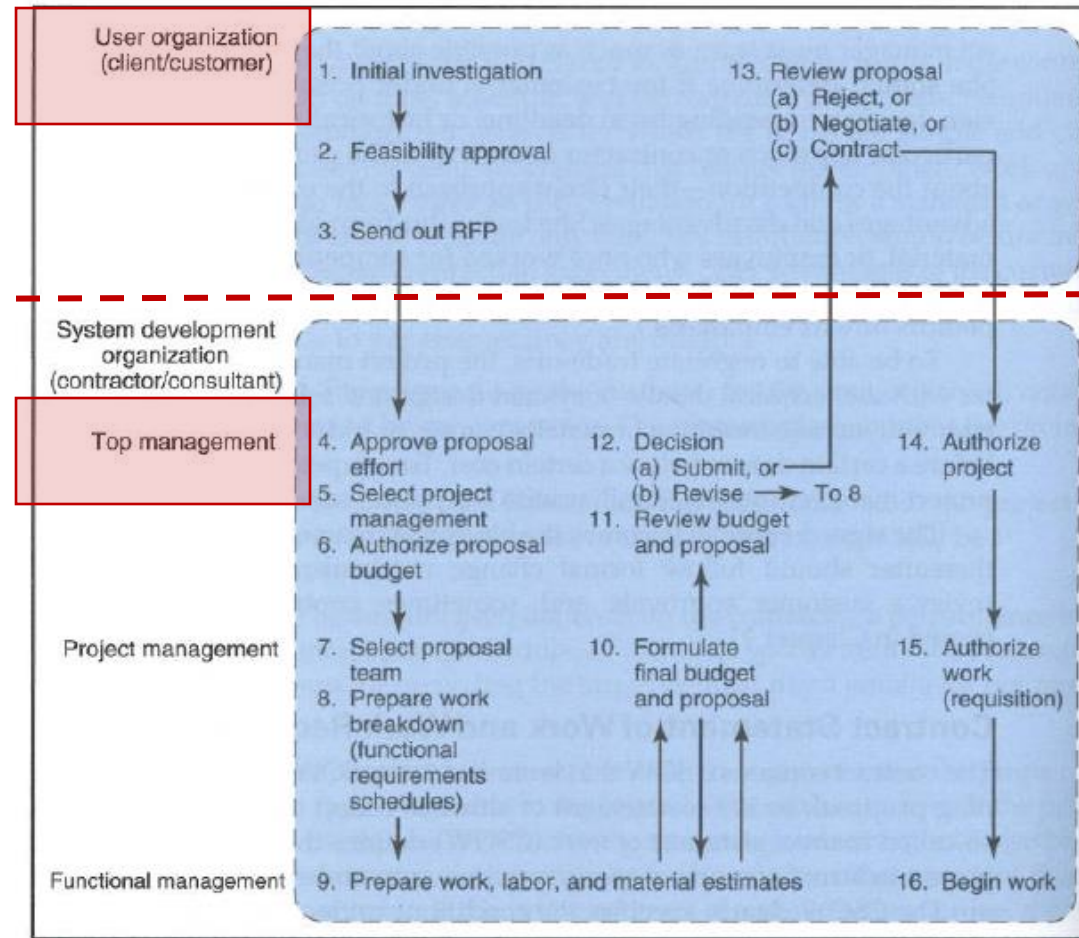
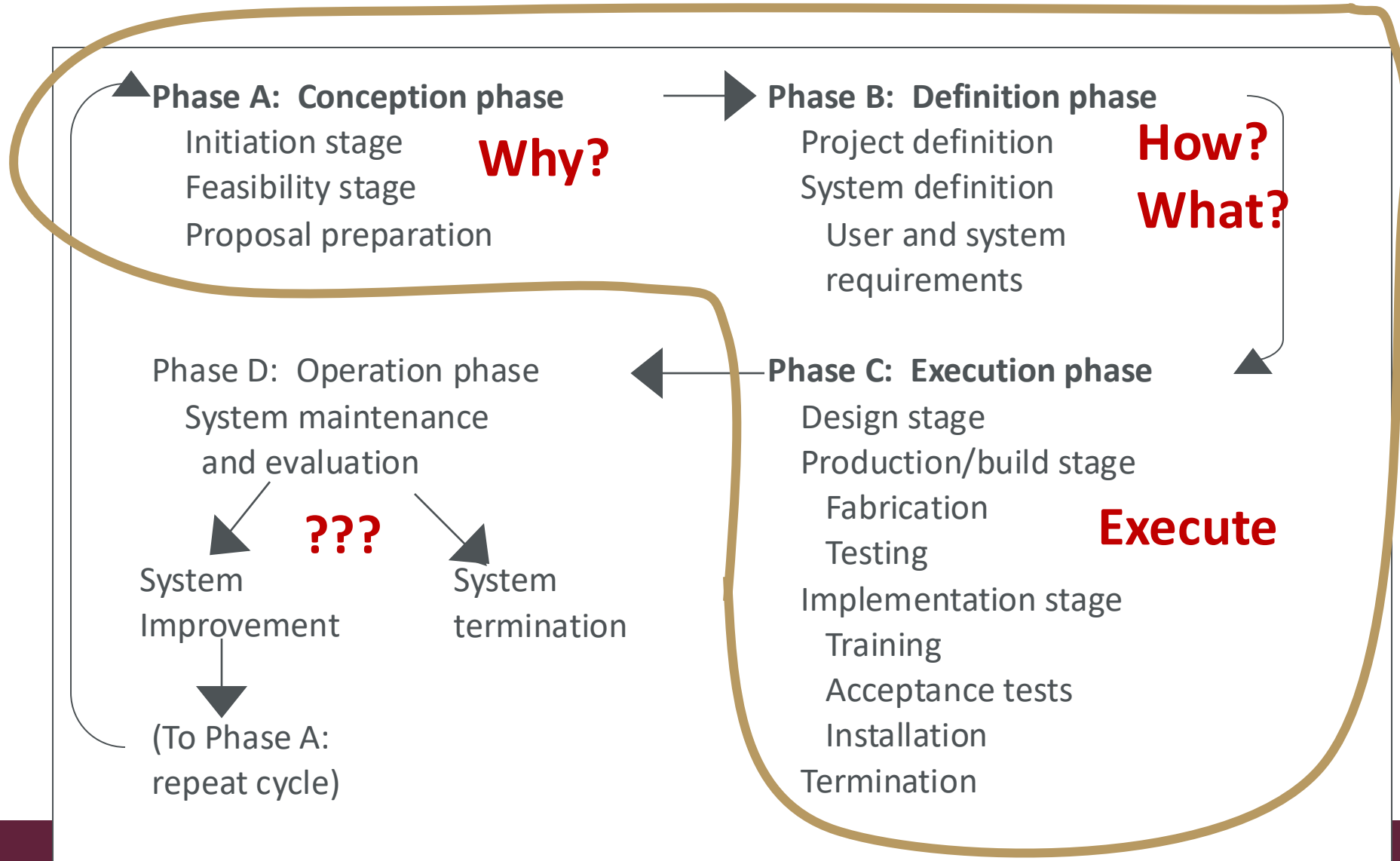


Figure 3.8
Project initiation, proposal preparation, and project authorization process.

Phases A, B, C are “Project Life Cycle”



Systems Development Cycle: Phases



- Phase A: **Conception**
 - **Perceived** need or problem
 - Initial screening or **feasibility** study; Alternative solutions
 - Proposal
 - **Concept** approval/rejection

SDC Phases (cont'd)



- Phase B: **Definition** (“Birth”)
 - Specify **requirements** in detail:
 - User requirements
 - System requirements/ system specifications
 - Define project to produce end-item/deliver requirements:
 - **Project master plan**

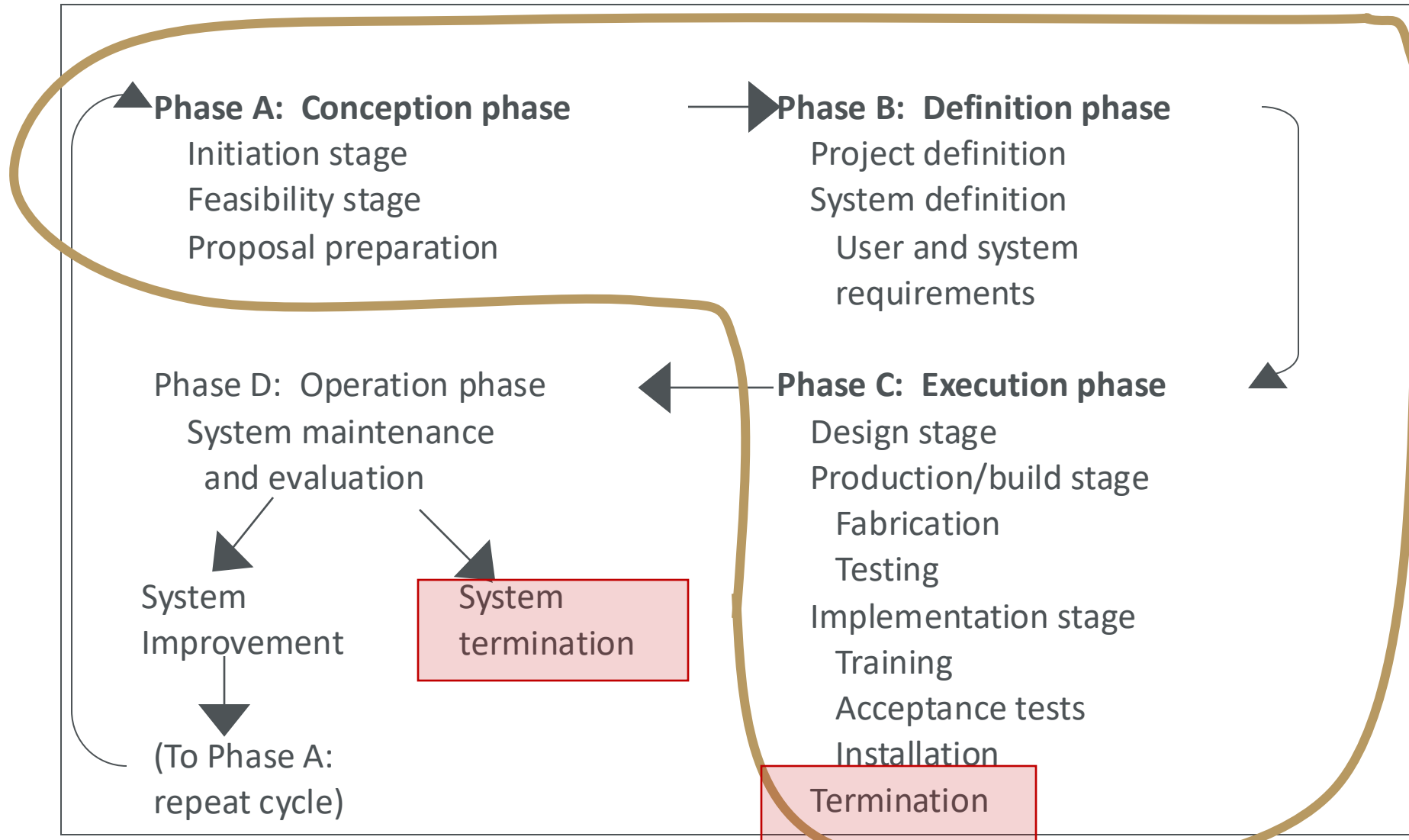
SDC Phases (cont'd)

- Phase C: **Execution** (“Growth”)
 - Design/development
 - Procurement/fabrication
 - Production/building
 - Installation

SDC Phases (cont'd)

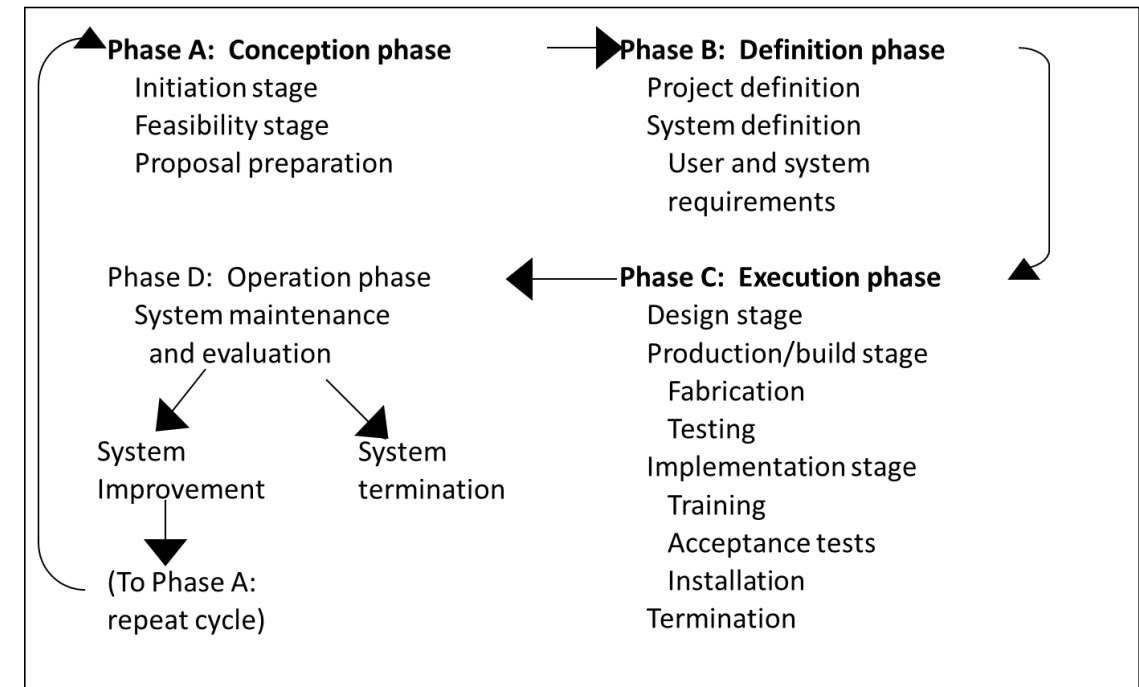
- Phase D: **Operation** (“Maturity”)
 - Customer gains **control**
 - System developer **might** remain involved with system/customer through:
 - Maintenance
 - Evaluation
 - Enhancement
 - Replacement

Phases A, B, C are “Project Life Cycle”



Project Life Span, Variations

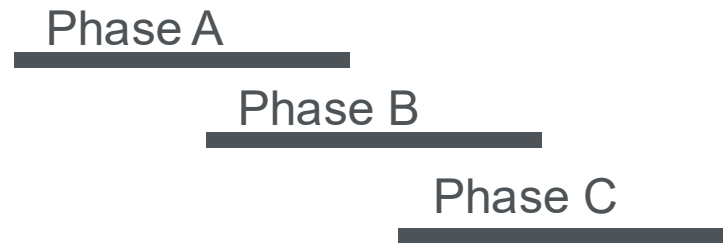
- **Phased** Project/Program Planning
 - In very large projects, phases A, B, and C are treated somewhat **independently**, almost like separate projects
 - Each phase **requires justification and approval**.
 - Project can be **cancelled** in each phase.
 - Different **contractors** can be involved in each phase.
 - Examples: phase/gate methodology and DOD development/production contracting



Project Life Span, Variations (cont'd)



- Fast-tracking
 - Phases (and stages within phases) **overlap** so work in successive phases (stages) happens simultaneously.



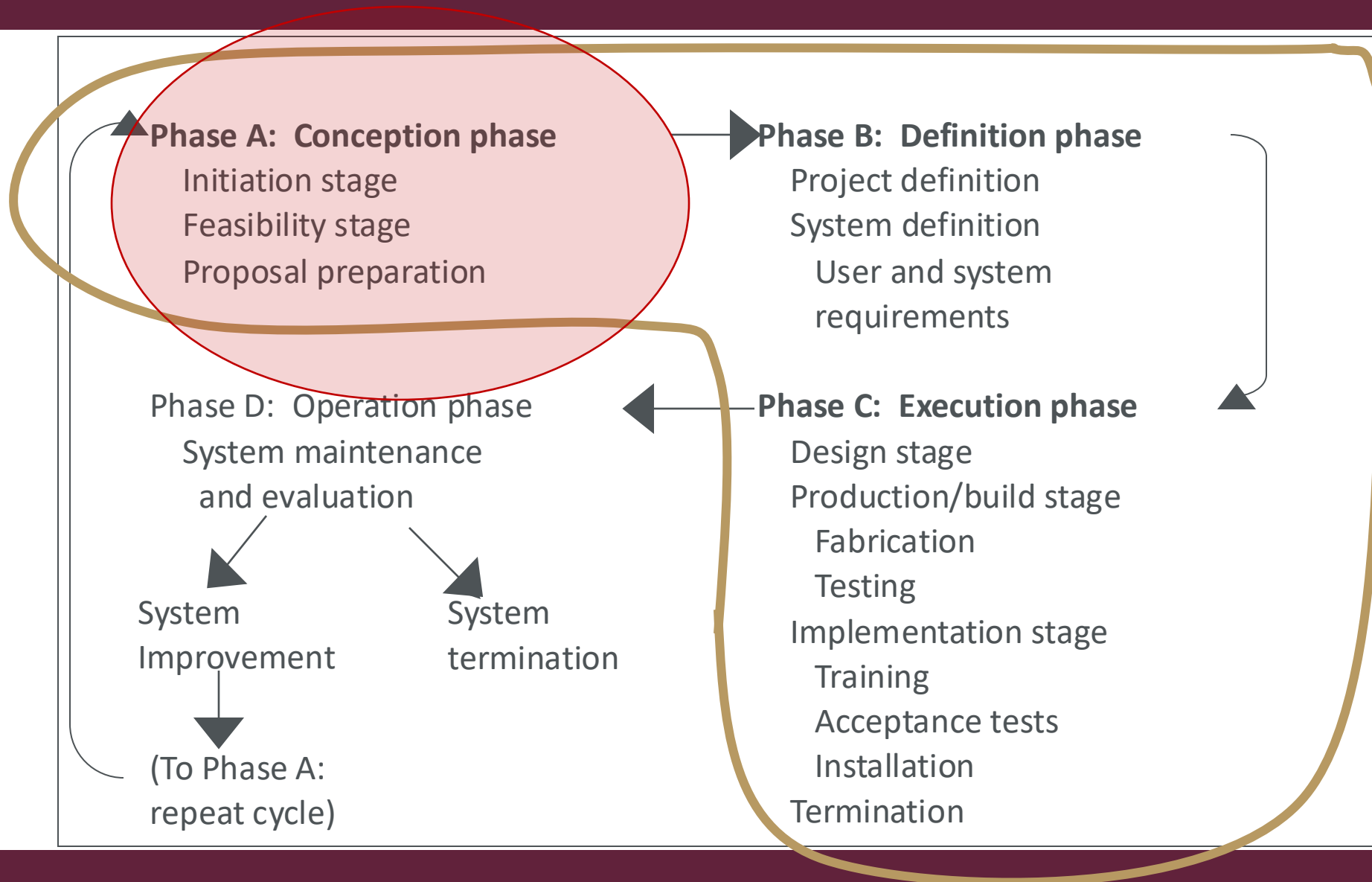
- Pro: **shorten** project life span
- Con: Increase **risk** of mistakes/ rework

3.3 Phase A: Project Conception

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Phases A, B, C are “Project Life Cycle”



Phase A: Conception



- Project Conception involves:
 - Initiation
 - RFPs
 - Feasibility
 - Needs Analysis
 - Proposal preparation
 - Proposal evaluation/project selection
 - Contracting

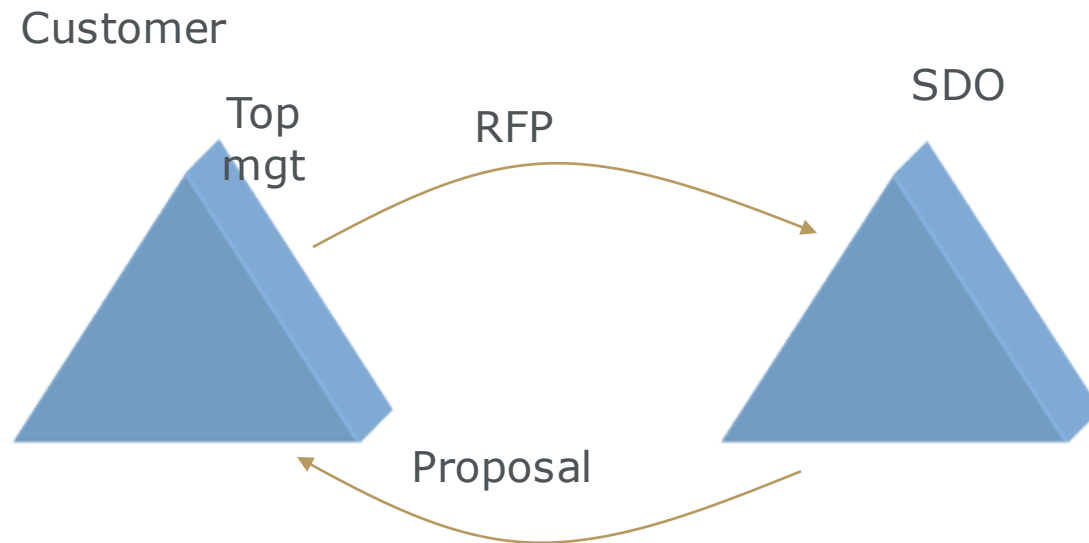
Initiation

- Every project starts in **somebody's mind** as a need or problem, or as a solution to somebody else's need or problem.
- An “**idea**” develops based on perceived problem, opportunity, or need
- Idea can originate **anywhere** in customer or contractor organizations
- To determine if idea has merit, instigate an initial (usually brief) investigation
 - Data sources
 - Interviews
 - Background research
 - Documentation

Usually
customer
does this

RFP (Request for Proposal)

- If customer decides to proceed with idea, next step is to **contact** an SDO (developer or contractor) to
 - investigate idea further, **OR**
 - do the work and deliver the solution/end-item
- **Contact** between customer and SDO **initiated** with the RFP
 - Also called RFB, RFQ, or RFT (bid, quote, or tender)



Example RFP

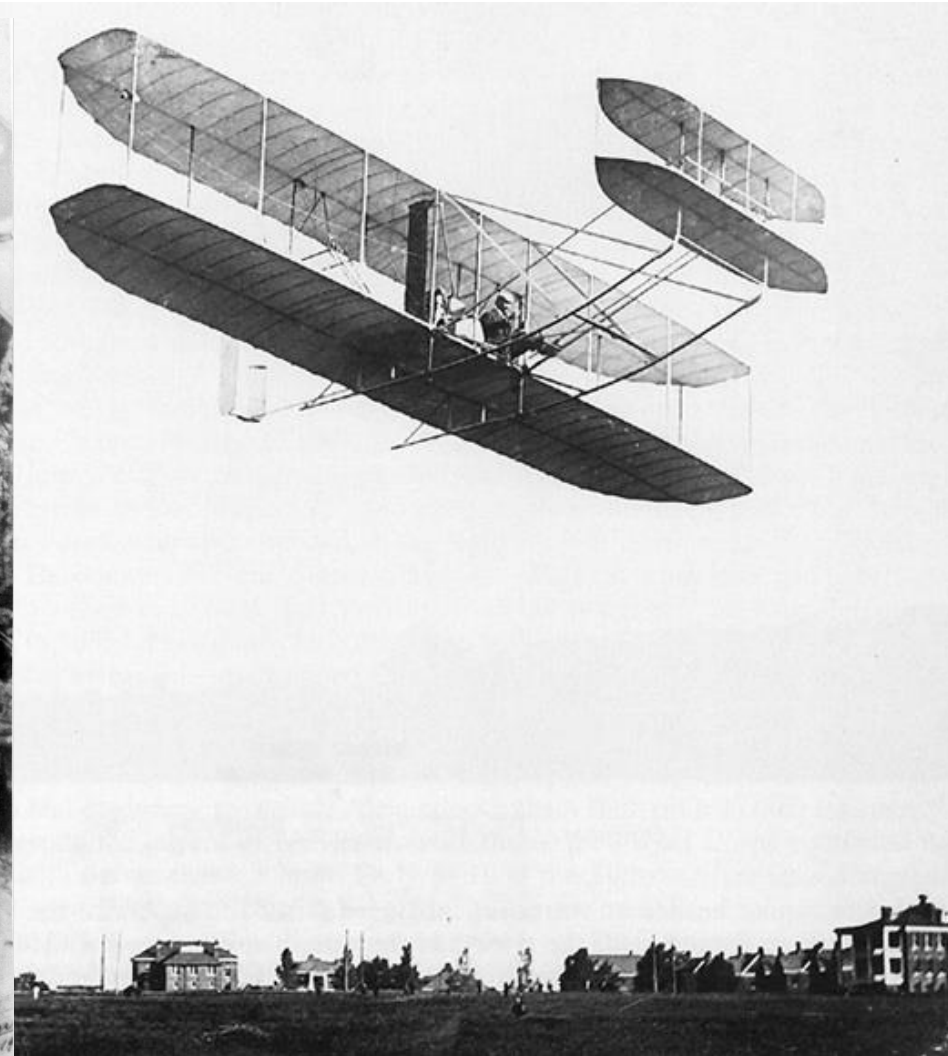
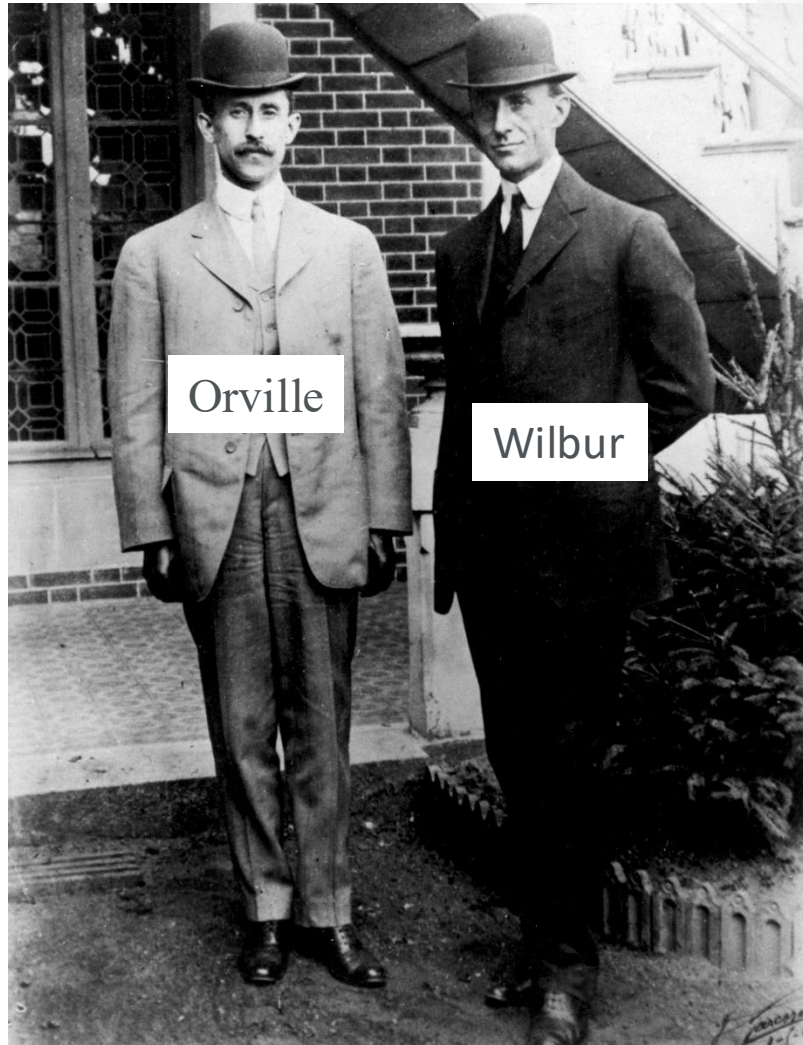
- Requirements for “Machine”:
 - Will be tested in front of Army officers
 - Must be able to carry for one hour a passenger besides pilot, the two of them weighing not less than 350 lbs.
 - Must show average speed of 40 miles per hour in ten-mile test
 - Must carry enough fuel for 125 miles
 - Must have “demountability” (built in such a way it can be taken apart and later reassembled without too much difficulty, so it can be transported on an army truck)
 - Must submit with bid 10 percent of price of machine as sign of good faith!

Example RFP (cont'd)



- Answer:
 - This is a Request for Bid from U.S. Army for “Flying Machine,” Nov. 1907
- Responses received: Forty-one!
 - Only three included required 10 percent deposit; two of these drop out
 - Only one bidder was actually capable of meeting the requirements...

The Wright Brothers



RFP Contents



Statement of Work (SOW)

- Description/background of problem, need, or general type solutions to be investigated
- Scope of work to be performed
- **Requirements** for results or end-item;

Proposal Requirements

- Conditions placed on proposal
 - proposal contents and format
 - data requirements
 - sample forms to include
 - submission location and **deadline**
- All proposals should “look” same

RFP Contents



Contractual Provisions

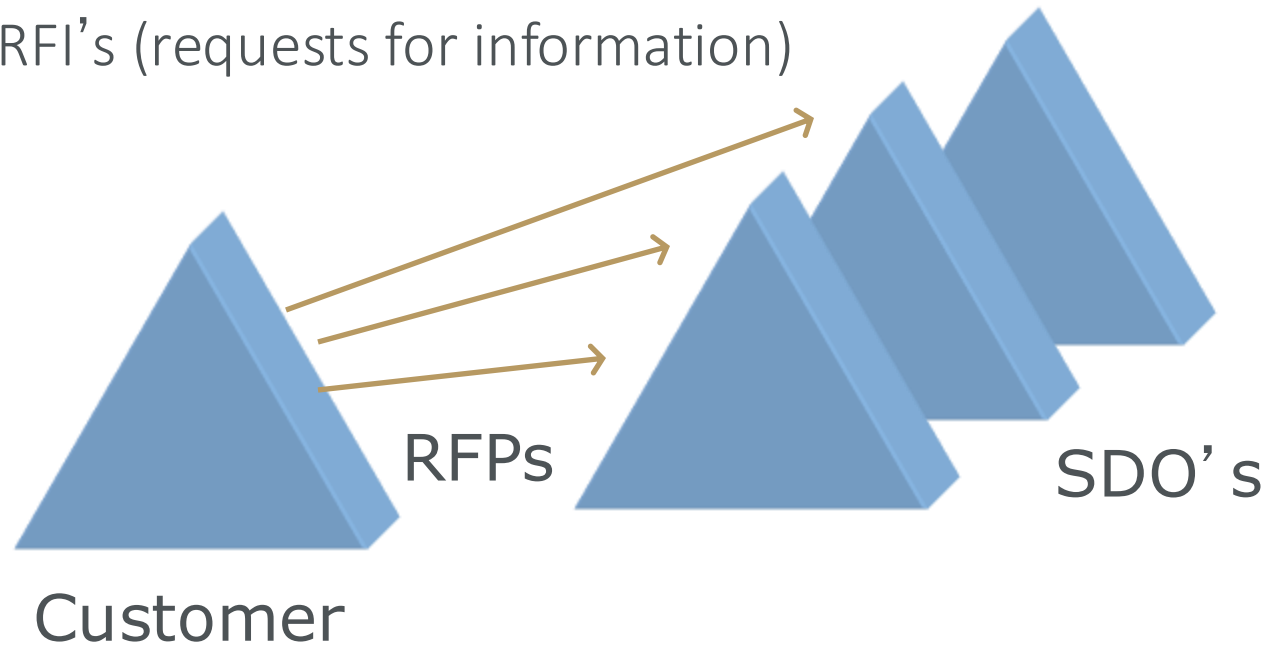
- Type of contract to be awarded and contractual provisions
 - Fixed-price
 - Cost-plus
 - Incentives
 - Special considerations

Additional Information or Data

- Name of contact person for requesting additional data—as necessary to enable SDO develop solution and prepare proposal or price quote
- Technical information to support the SOW

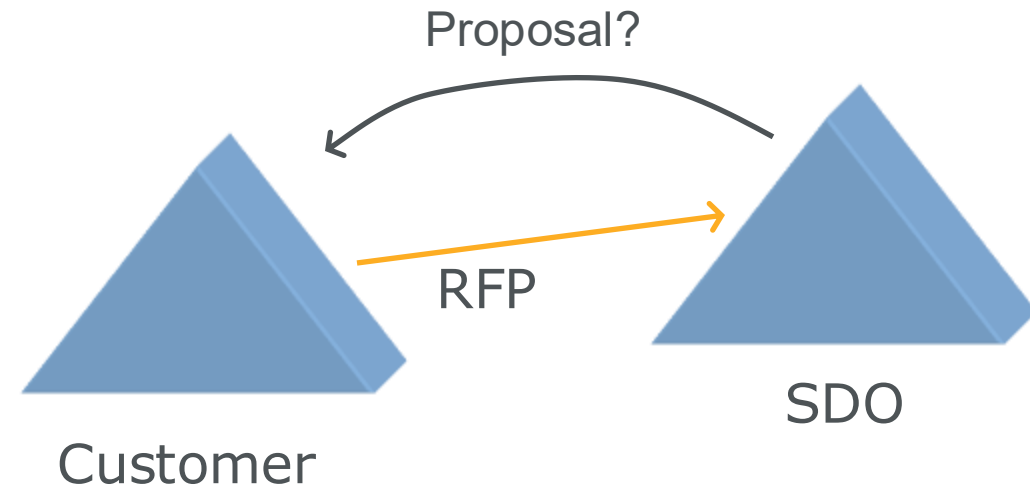
RFP Process

- Customer **sends RFP's** to
 - SDO's on bidder's list
 - SDO's recommended by other customers
 - SDO's requesting an RFP (as advertised in Commerce Business Daily or trade newsletters)
 - Qualified SDO's based upon RFI's (requests for information)



SDO Decision to Respond to RFP

- SDO **considers** whether
 - **Competition** has head start
 - SDO has **funds**, facilities, etc., to devote to project
 - **Value** of project in terms of work reputation, profitability, etc., is worth it
 - Contract **can be won**
 - Project is **doable** for SDO



3.4 Project Feasibility

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Feasibility Study



- Most projects are **preceded** by detailed study/analysis to compare costs with benefits/outcomes
- Study considers “**feasibility**” of idea or proposed work given proposed benefits constrained resources
- Feasibility study might be **done by customer** during initial investigation...

Feasibility Study (cont'd)

- Alternatively, might be **done by** SDO
- Regardless, somewhere in the Conception Phase a feasibility study is performed to **justify** the “idea” or proposed solution

Feasibility Study



- Sometimes, outcome of study is a “Business Case” to convince customer/sponsor to authorize and undertake the project. Typical contents:
 - Cost-benefit analysis: estimated project costs compared to the benefits
 - Estimated project duration when financial return depends on the timescale
 - Financial aspects such as the funding approach
 - Risks, issues, and a preliminary risk management plan
 - Assumptions

Environmental Impact



In some projects, part of the feasibility is to assess the *Environmental Impact* (EI) of the project.

Typical contents of an EI assessment and report include

- Summary of proposed development and/or management plans.
- Alternative sites and technologies to the proposed project
- Description of project's existing site and surrounding area
- Potential project impacts, such as on
 - Quality of air, soil, watersheds, wetlands, flood plains
 - Fisheries; sensitive plants; sensitive, endangered, or threatened species
 - Scenic resources; societal and aesthetic experiences
 - Heritage resources (sites, structures, buildings, districts, objects)
 - Historical resources (logging, ranching, grazing, mining, recreation)
- Adverse impacts that cannot be avoided
- Long term impacts on resources.
- Ways to prevent, minimize or offset impacts; ways to monitor actual impacts

Sustainability



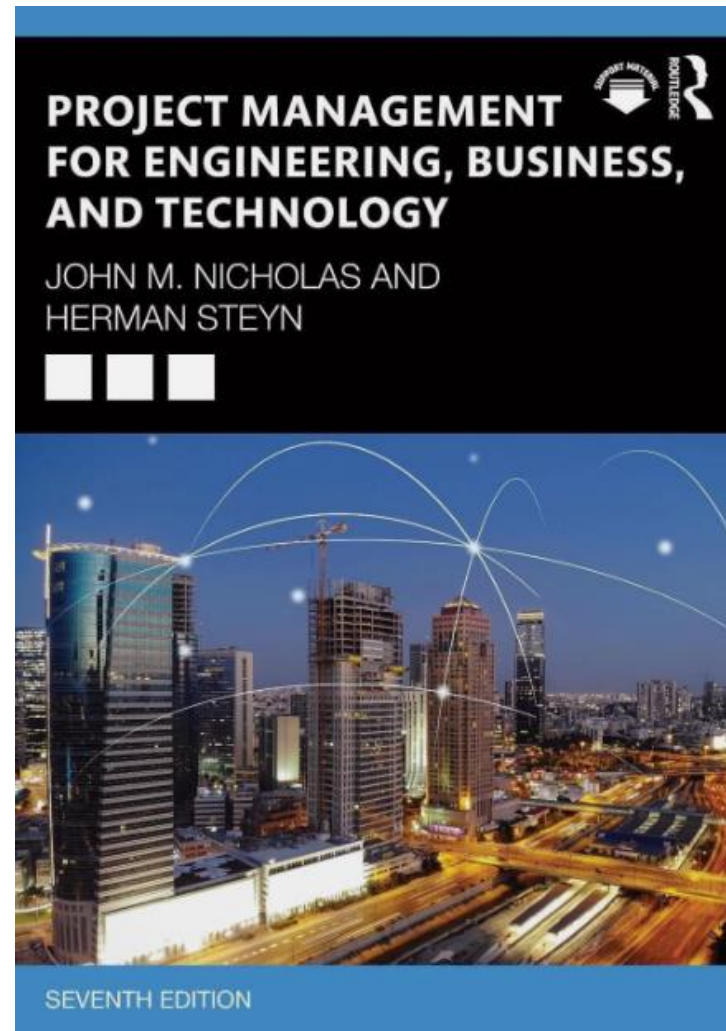
Many industries have taken strides to incorporate *environmental and social responsibility* into the role of project management. The feasibility study might include guidelines or considerations. Examples in construction include guidelines for:

- Passive ventilation systems
- Whole building heat recovery systems
- Renewable/recycled materials
- Reduce landfill waste: crush/reuse aggregates (stone, etc.)
- Use suppliers who accept returns/exchanges, reuse packaging, and use reclaimed/recycled building products
- Minimize dust from concrete/mortar; avoid air/water pollution

The study should ensure that the project and its outcomes comply with local, state, and federal environmental laws.

3.5 The Project Proposal

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Proposals



- The proposal is one of **most important** documents in project life span.
- Purposes:
 1. **Sales**
 - SDO describes to customer **what** it will do
 - Proposal must “**sell**” customer
 - Getting contract depends on proposal
 2. **Contract**
 - Customer assumes SDO will **be able to deliver** on what is said in proposal
 - If both customer and SDO **sign off** on proposal, contract is created to deliver **whatever** SDO proposed

Contents of Proposal



- Standard Contents
 - Executive Summary
 - Technical Section (Statement of Work)
 - Cost and Payment Section
 - Legal Section
 - Management/Qualifications Section

Customer Review of Proposal



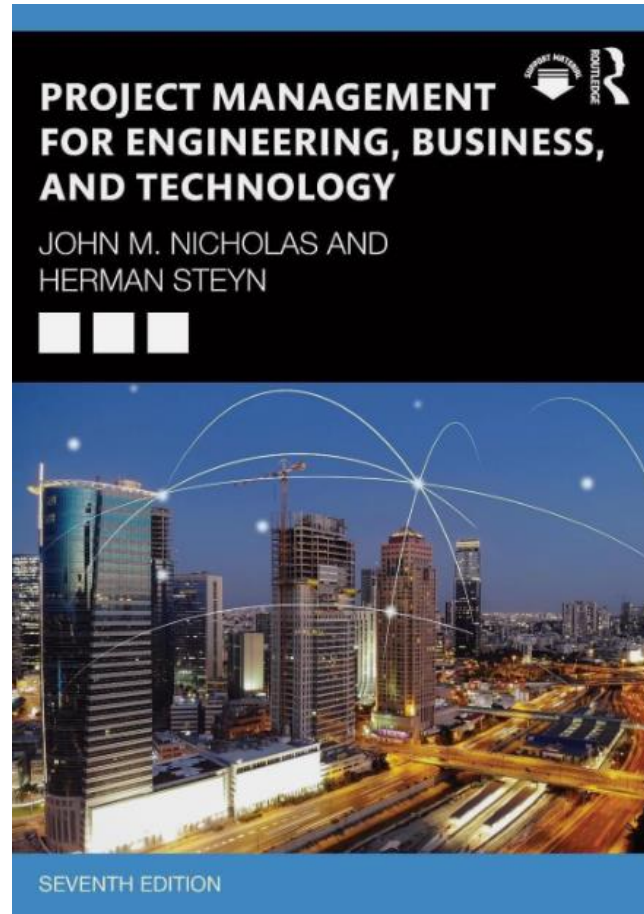
- Customer **evaluates**
 - Cost
 - Benefits
 - Likelihood of Success
 - Contractor Reputation
 - Narrow the List
- Often a **handful** of good proposals remain after many others have been discarded. These go to **negotiation**.

Negotiation



- Customer and SDO **meet** to
 - **Clarify** terms (ensure common understanding)
 - **Reach Agreement** on requirements, schedule, or price

3.6 & 3.7 Project Contracting



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Contracting



- Every project **involves** contracting
 - an agreement for one party (SDO) to do something (project) for another (customer)
- Most people think of project contract **only** between customer and SDO
 - in many projects the SDO is also a customer that **contracts its work** to many other organizations

Basic Kinds of Contractual Agreements



Different contractual agreements offer different **advantages** to the customer and contractor, depending on the nature of the project.

(These are **discussed in Chapter 12**)

The basic agreements are:

- **Fixed Price Contract**—Price paid by the customer for the project is fixed regardless of the costs incurred by the contractor.
- **Cost-Plus Contract**—Price paid by the customer is based on the costs incurred in the project plus the contractor's fee.
- **Incentive Contract**—Price paid by the customer depends on the contractor's performance in comparison to the target price, schedule, or technical specification: the contractor either receives a bonus for exceeding the target, or must pay the customer a penalty for falling short of the target.

Process: RFP, Proposal and Contract

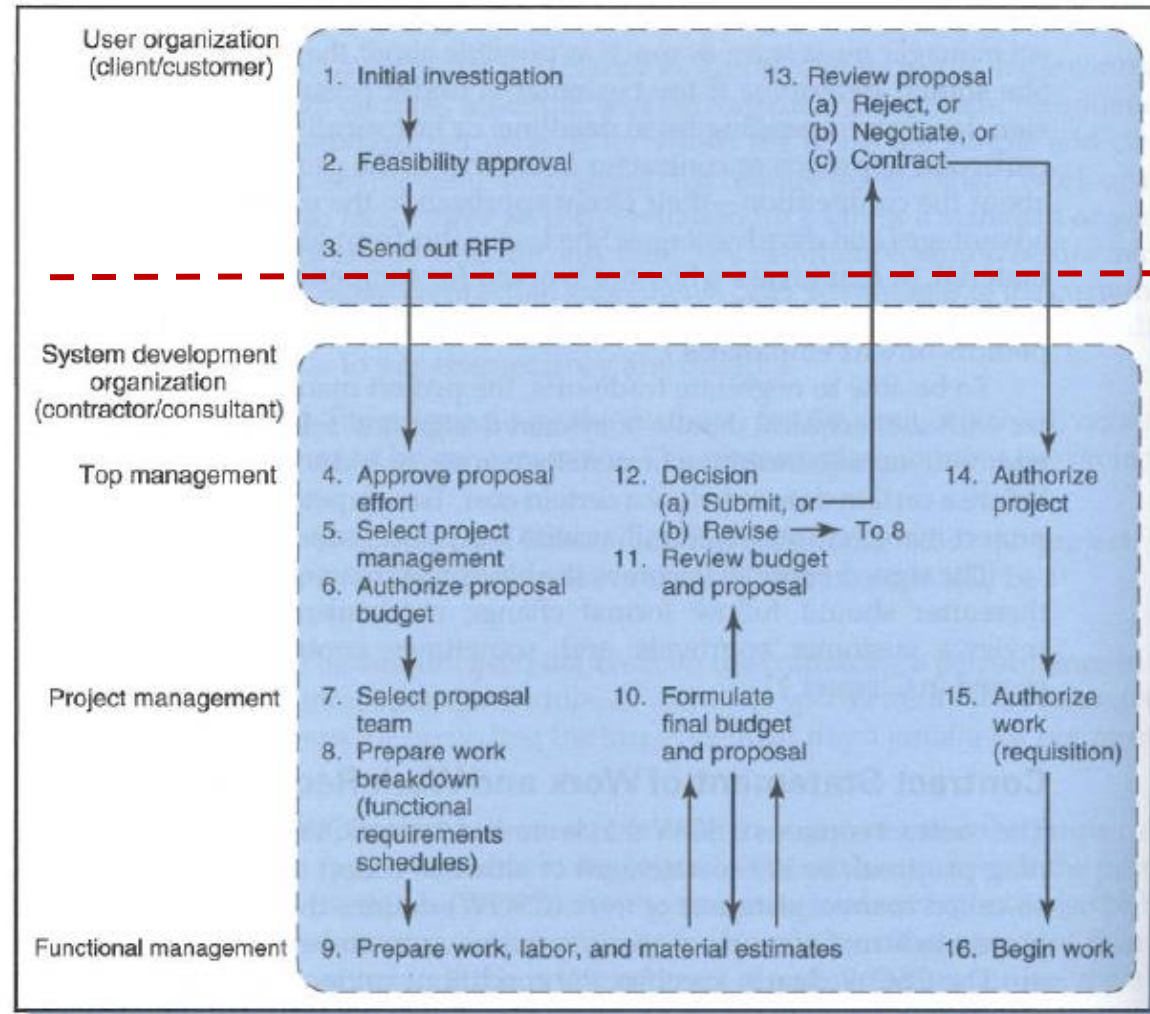


Fig 3.9

Thank you
Enkosi
Dankie

